case O(h)=O(n), for skewed trees.

· We omitted the space analysis, because all these methods only require constant extra space.

Table 3.2 Operations provided by treaps, and their cost assuming n elements and height h

Operation	Running Time	Worst Case
Insert	O(h)	O(n)
Top	O(h)	O(n)
Remove	O(h)	O(n)
Peek	O(1)	O(1)
Contains	O(h)	O(h)
UpdatePriority	O(h)	O(n)
Min/Max	O(h)	O(n)

3.4 Applications: Randomized Treaps

So, we are now able to implement our inventory, keep track of the products in stock, and extract the ones closest to running out of stock: that would certainly impress everyone at the next family reunion!

Hopefully our example helped you understand how treaps work, but… I have a confession to make: treaps are not really used as a way to index multidimensional data.

We'll see in the next chapters, and in particular in chapter 7, when we talk about cache, that there are better ways to address problems equivalent to the example we presented in this chapter.